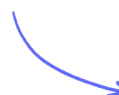


Congruence & Similarity

Congruence / Similarity / Similar Lengths / Similar Areas & Volumes

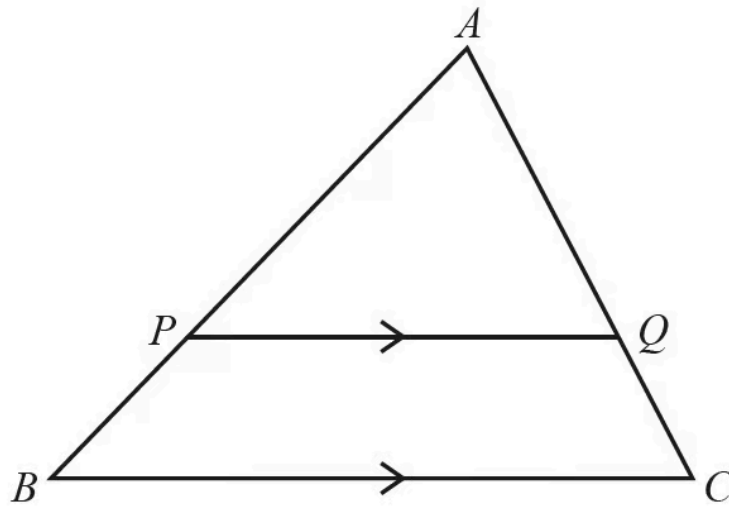
Medium (10 questions)	/28
Hard (6 questions)	/16
Very Hard (5 questions)	/25
Total Marks	/69

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Medium Questions

1



NOT TO
SCALE

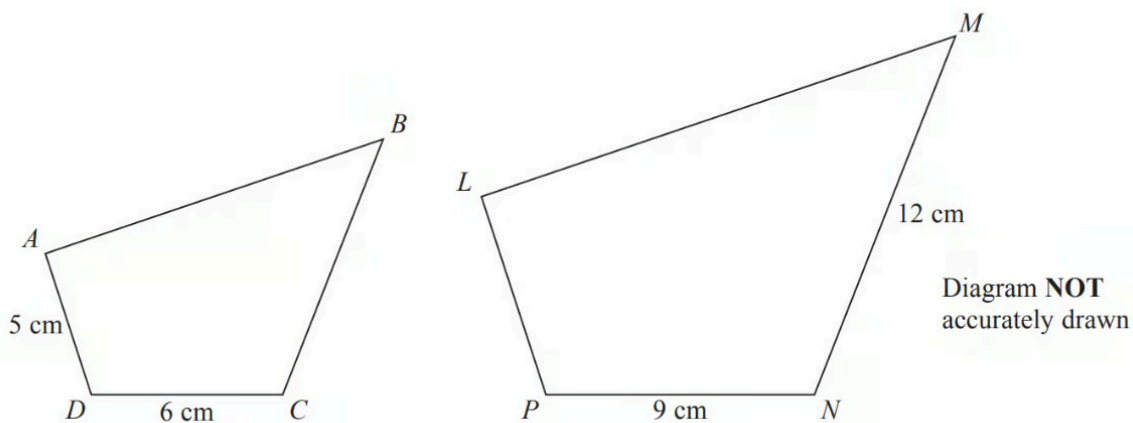
In the diagram, PQ is parallel to BC .
 APB and AQC are straight lines.
 $PQ = 8$ cm, $BC = 10$ cm and $AB = 9$ cm.

Calculate PB .

$PB = \dots\dots\dots$ cm

(2 marks)

2 (a)



Quadrilaterals $ABCD$ and $LMNP$ are mathematically similar.

Angle A = angle L

Angle B = angle M

Angle C = angle N

Angle D = angle P

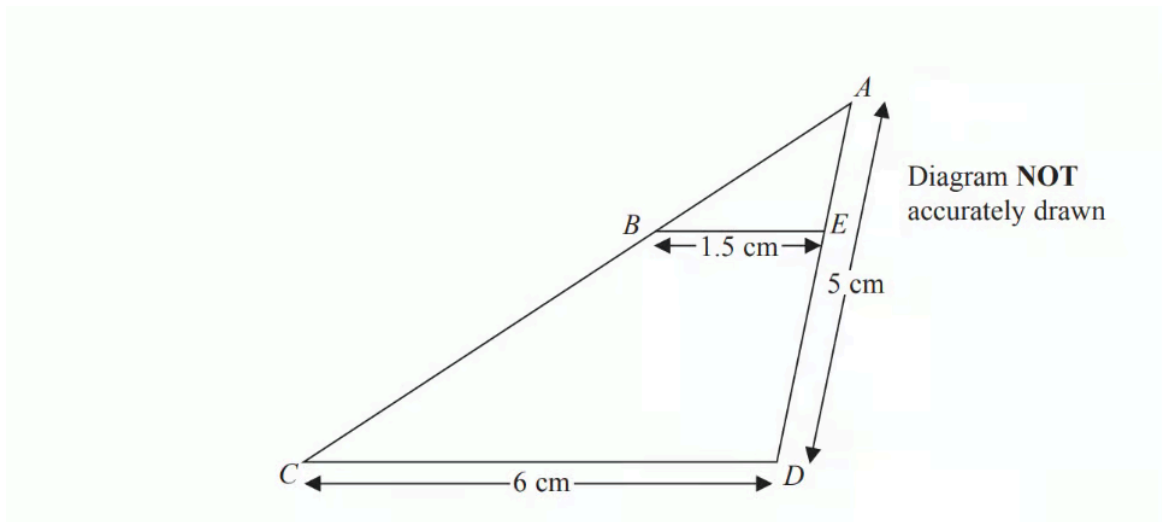
Work out the length of LP .

(2 marks)

(b) Work out the length of BC .

(2 marks)

3



ABC and AED are straight lines. BE and CD are parallel.

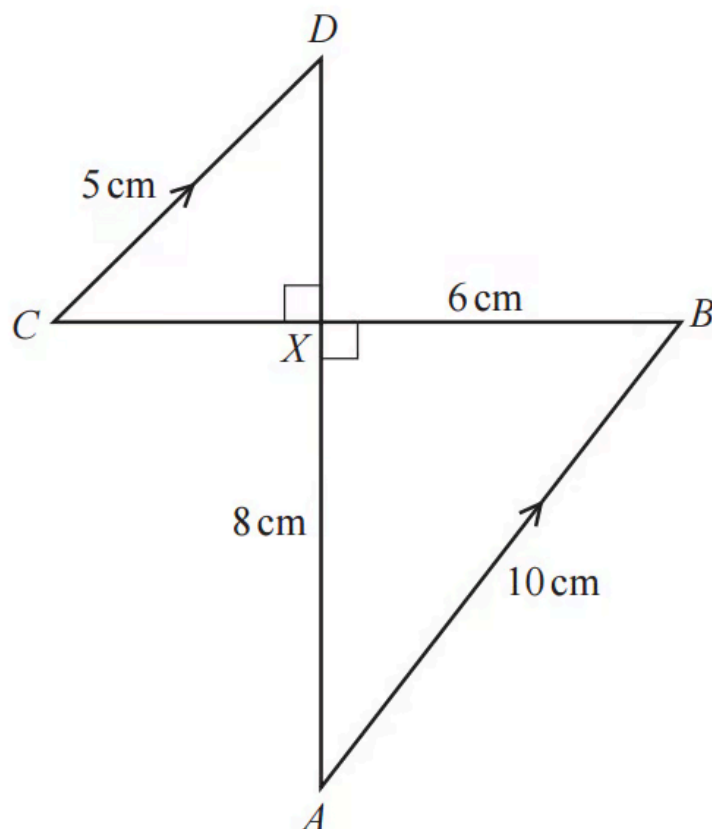
$BE = 1.5$ cm.

$CD = 6$ cm.

$AD = 5$ cm.

Calculate the length of ED .

(3 marks)



NOT TO
SCALE

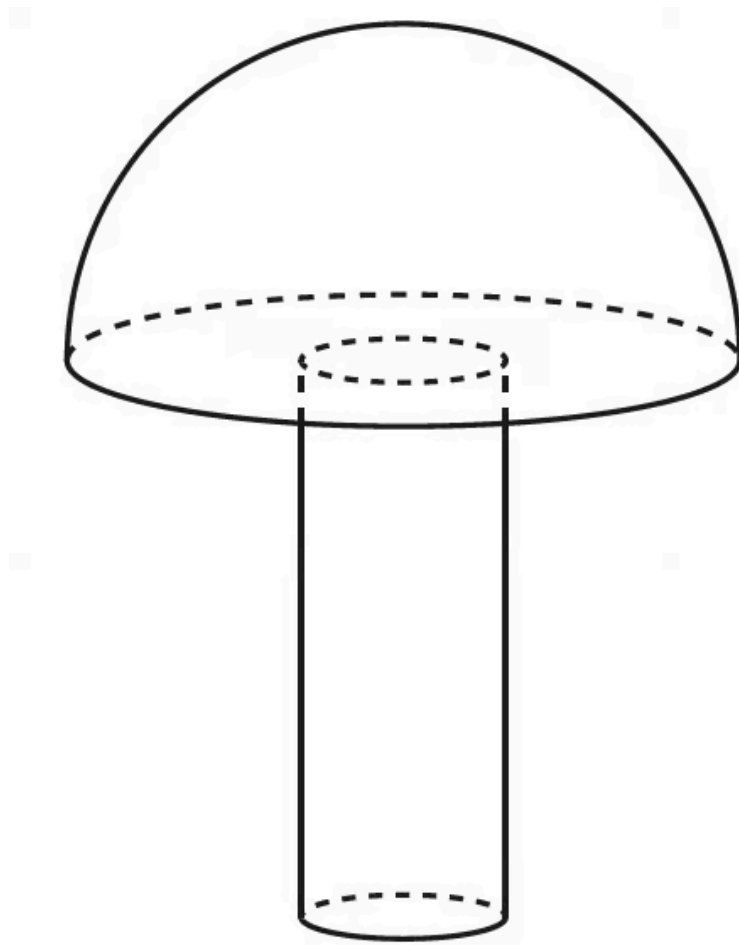
In the diagram, AB and CD are parallel.
 AD and BC intersect at right angles at the point X .
 $AB = 10$ cm, $CD = 5$ cm, $AX = 8$ cm and $BX = 6$ cm.

Use similar triangles to calculate DX .

$DX = \dots\dots\dots$ cm

(2 marks)

- 5 A solid cylinder with radius 1.6 cm is attached to the hemisphere to make a toy.



NOT TO
SCALE

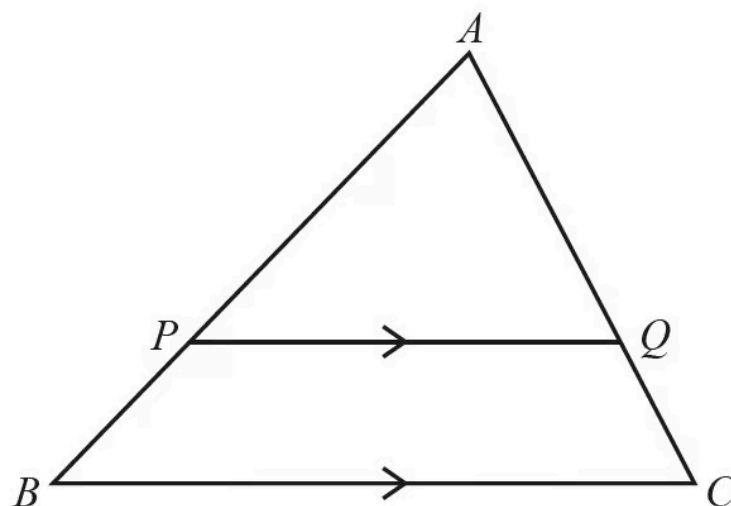
The total volume of the toy is 300 cm^3 .

A mathematically similar toy has volume $19\,200 \text{ cm}^3$.

Calculate the radius of the cylinder for this toy.

..... cm

(3 marks)



NOT TO
SCALE

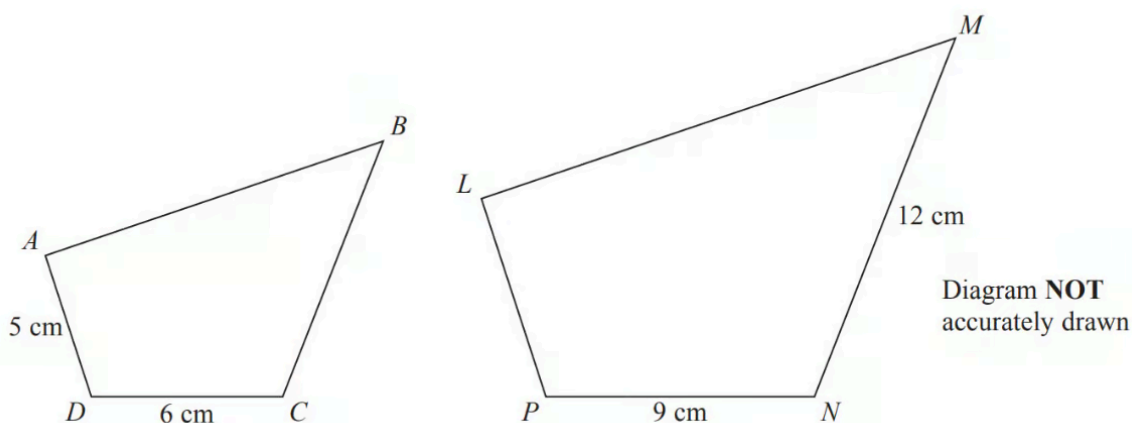
In the diagram, PQ is parallel to BC .
 APB and AQC are straight lines.
 $PQ = 8$ cm, $BC = 10$ cm and $AB = 9$ cm.

Calculate PB .

$PB = \dots\dots\dots$ cm

(2 marks)

7 (a)



Quadrilaterals $ABCD$ and $LMNP$ are mathematically similar.

Angle $A = \text{angle } L$
 Angle $B = \text{angle } M$
 Angle $C = \text{angle } N$
 Angle $D = \text{angle } P$

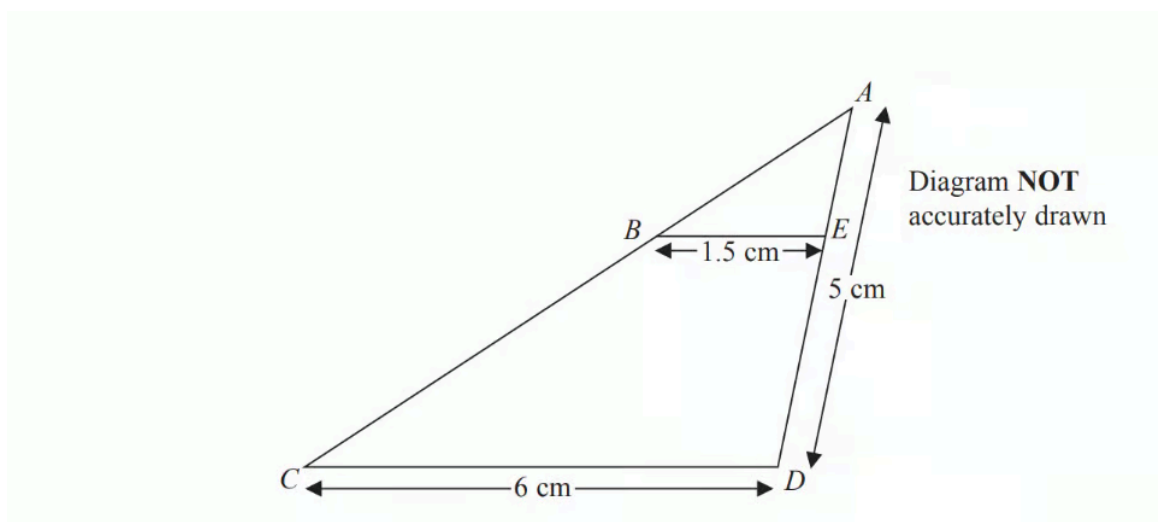
Work out the length of LP .

(2 marks)

(b) Work out the length of BC .

(2 marks)

8



ABC and AED are straight lines. BE and CD are parallel.

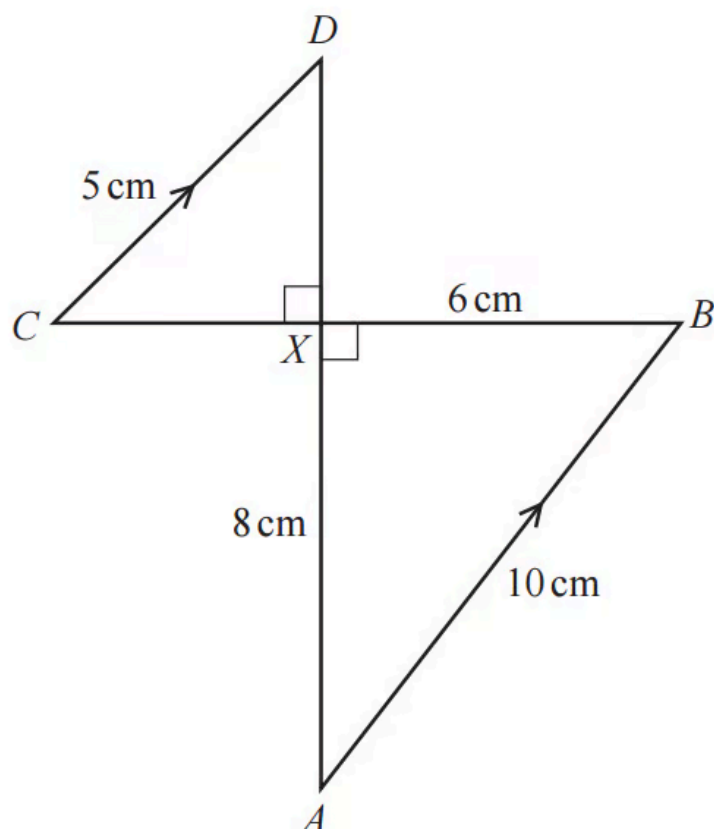
$BE = 1.5 \text{ cm}$.

$CD = 6 \text{ cm}$.

$AD = 5 \text{ cm}$.

Calculate the length of ED .

9

NOT TO
SCALE

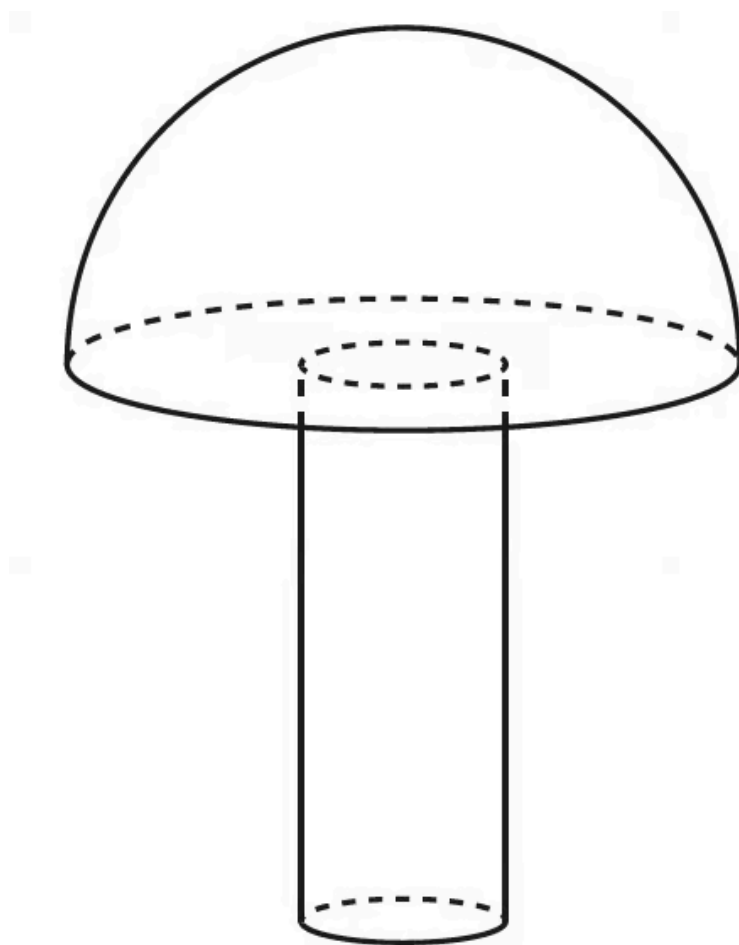
In the diagram, AB and CD are parallel.
 AD and BC intersect at right angles at the point X .
 $AB = 10$ cm, $CD = 5$ cm, $AX = 8$ cm and $BX = 6$ cm.

Use similar triangles to calculate DX .

$DX = \dots\dots\dots$ cm

(2 marks)

- 10 A solid cylinder with radius 1.6 cm is attached to the hemisphere to make a toy.



NOT TO
SCALE

The total volume of the toy is 300 cm^3 .

A mathematically similar toy has volume $19\,200 \text{ cm}^3$.

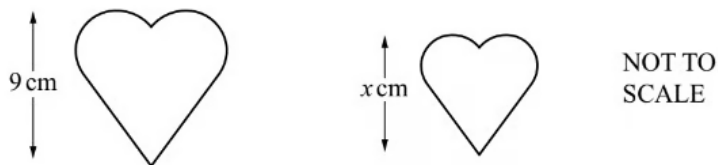
Calculate the radius of the cylinder for this toy.

..... cm

(3 marks)

Hard Questions

1



The two shapes are mathematically similar.

The area of the larger shape is 36 cm^2 and the area of the smaller shape is 25 cm^2 .

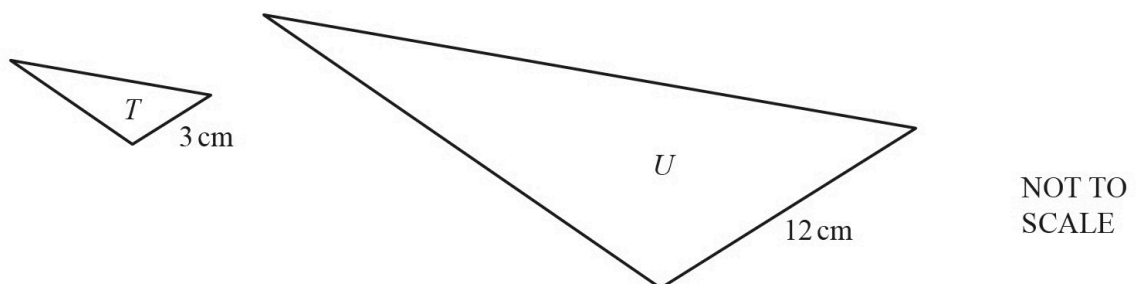
The height of the larger shape is 9 cm and the height of the smaller shape is x cm.

Find the value of x .

$x = \dots\dots\dots$

(3 marks)

2



The diagram shows two mathematically similar triangles, T and U .

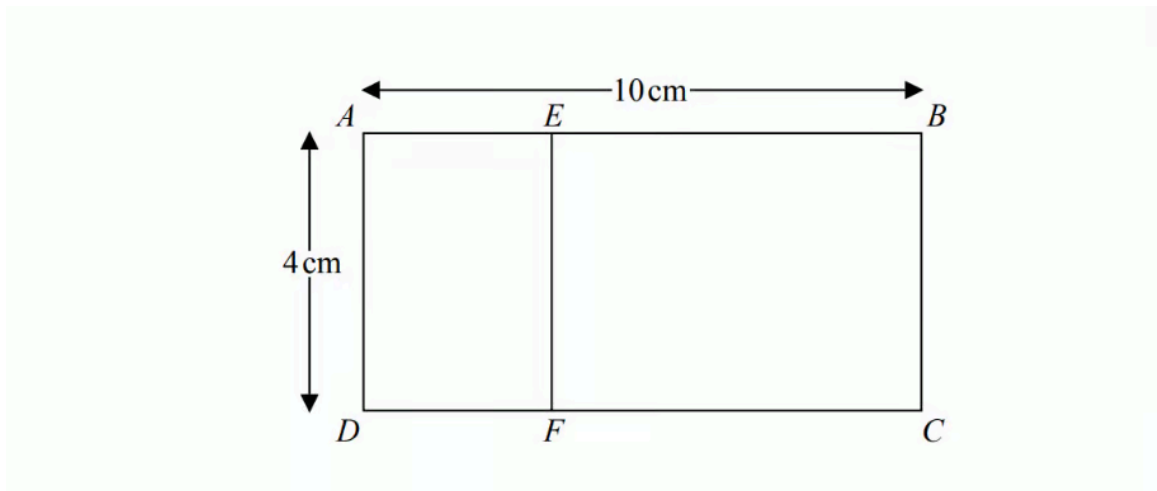
Two corresponding side lengths are 3 cm and 12 cm. The area of triangle T is 5 cm^2 .

Find the area of triangle U .

$\dots\dots\dots \text{ cm}^2$

(2 marks)

- 3 Rectangle $ABCD$ is mathematically similar to rectangle $DAEF$.



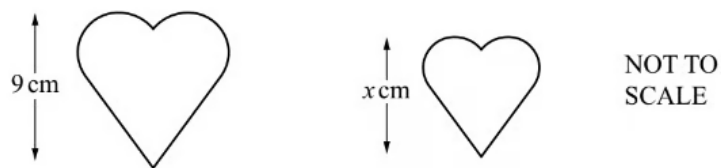
$$AB = 10\text{ cm.}$$

$$AD = 4\text{ cm.}$$

Work out the area of rectangle $DAEF$.

(3 marks)

4



The two shapes are mathematically similar.

The area of the larger shape is 36 cm^2 and the area of the smaller shape is 25 cm^2 .

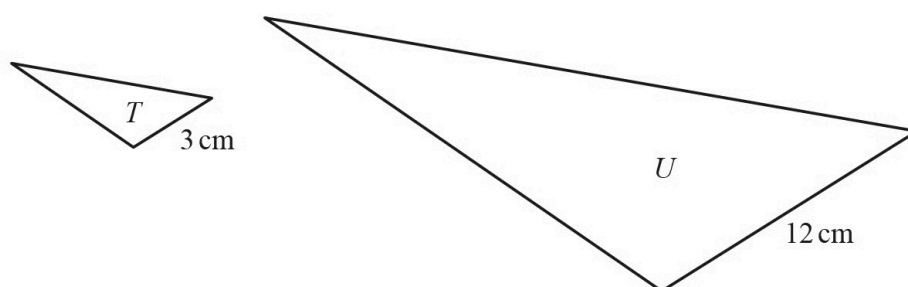
The height of the larger shape is 9 cm and the height of the smaller shape is $x\text{ cm}$.

Find the value of x .

$x = \dots\dots\dots$

(3 marks)

5



NOT TO
SCALE

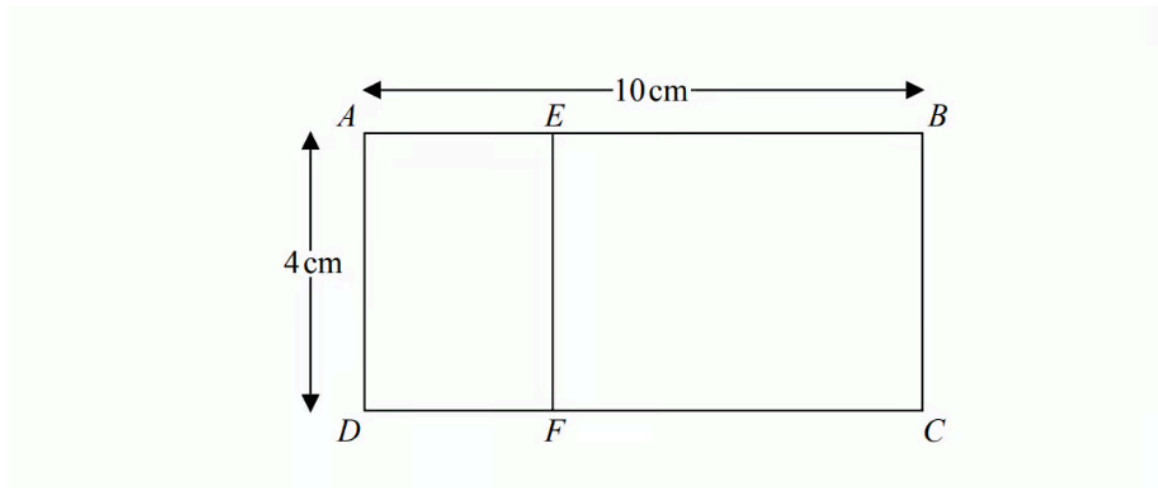
The diagram shows two mathematically similar triangles, T and U .
Two corresponding side lengths are 3 cm and 12 cm. The area of triangle T is 5 cm^2 .

Find the area of triangle U .

$\dots\dots\dots \text{cm}^2$

(2 marks)

- 6 Rectangle $ABCD$ is mathematically similar to rectangle $DAEF$.



$$AB = 10 \text{ cm.}$$

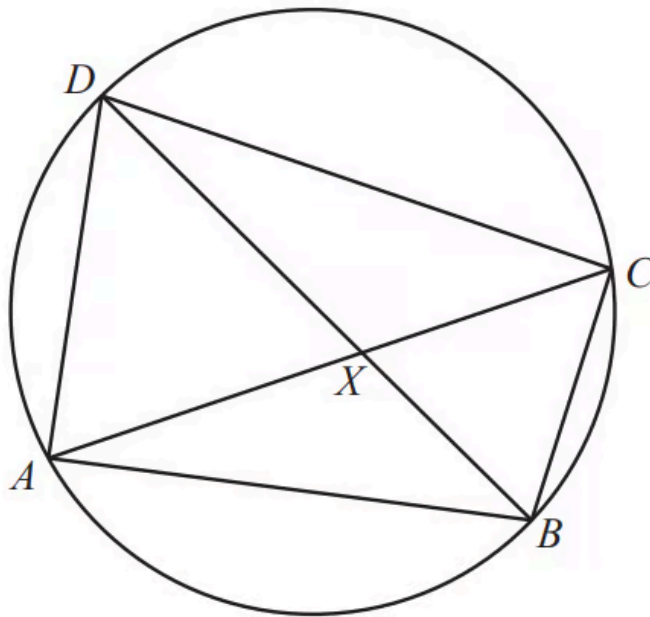
$$AD = 4 \text{ cm.}$$

Work out the area of rectangle $DAEF$.

(3 marks)

Very Hard Questions

1 (a)



NOT TO
SCALE

The diagonals of the cyclic quadrilateral $ABCD$ intersect at X .

Explain why triangle ADX is similar to triangle BCX . Give a reason for each statement you make.

(3 marks)

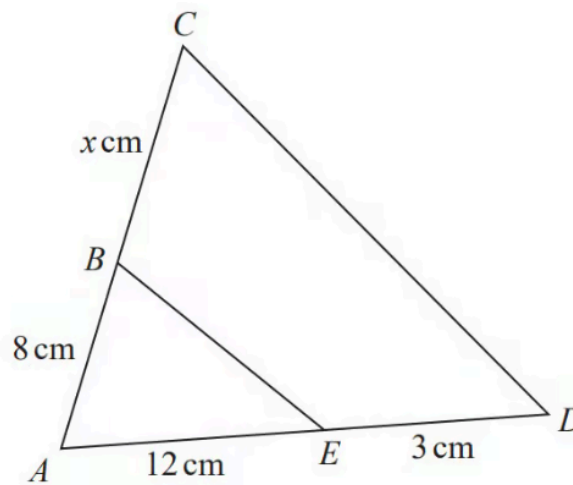
(b) $AD = 10$ cm, $BC = 8$ cm, $BX = 5$ cm, $CX = 7$ cm.

Calculate DX .

$DX = \dots\dots\dots$ cm

(2 marks)

- 2 The two triangles in the diagram are similar.



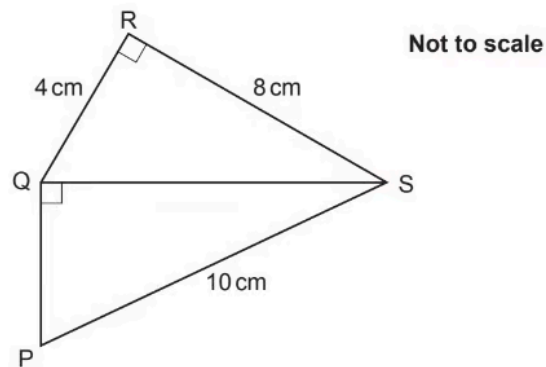
There are two possible values of x .

Work out each of these values.

State any assumptions you make in your working.

(5 marks)

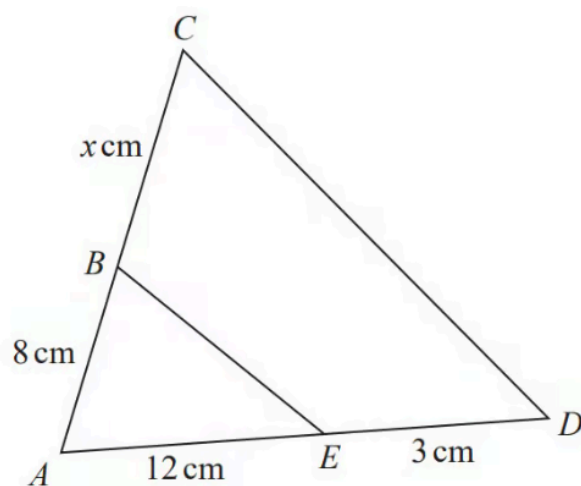
- 3 The diagram below shows two right-angled triangles.



Prove that triangles PQS and QRS are similar.

(5 marks)

- 4 The two triangles in the diagram are similar.

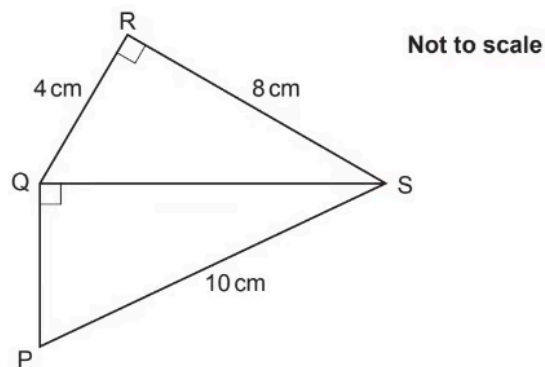


There are two possible values of x .

Work out each of these values.
State any assumptions you make in your working.

(5 marks)

- 5 The diagram below shows two right-angled triangles.



Prove that triangles PQS and QRS are similar.

(5 marks)